

Supplementary Material: Finsler Flow Matching for Amortized Time-Optimal Planning under Asymmetric Motion Constraints

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I. FINSLER PRELIMINARIES AND CLOSED FORMS

A. Randers metrics and Zermelo navigation

A Randers metric $F(q, v) = \sqrt{v^\top A v} + b^\top v$ with $\|b\|_{A^{-1}} < 1$ corresponds to navigation data (h, W) , a Riemannian metric h and a drift W with $\|W\|_h < 1$, through [1]

$$\lambda = 1 - \|W\|_h^2, \quad A = \frac{\lambda h + (hW)(hW)^\top}{\lambda^2}, \quad b = -\frac{hW}{\lambda}, \quad (1)$$

and the indicatrix $\{F = 1\} = \{W + e : \|e\|_h = 1\}$ is the unit h -ellipsoid displaced by W . The condition $\|W\|_h < 1$ states that the robot can always overcome the drift; beyond it the metric is of Kropina type, which we do not consider. The dual norm is again of Randers type,

$$F^*(q, \xi) = \xi^\top W + \|\xi\|_{h^{-1}}, \quad \mathcal{L}^{-1}(\xi) = F^*(\xi) \left(W + \frac{h^{-1}\xi}{\|\xi\|_{h^{-1}}} \right), \quad (2)$$

so the inverse Legendre transform needs no iteration. Geodesics are integrated in Hamiltonian form with $H = \frac{1}{2}F^*(q, p)^2$, $\dot{q} = \partial_p H = \mathcal{L}^{-1}(p)$, $\dot{p} = -\partial_q H$, using a fourth-order Runge–Kutta scheme; H is conserved to 10^{-9} along all trajectories used in the paper. In one dimension with $A = 1$, $b = \beta$ the forward and backward maximum speeds are $1/(1 + \beta)$ and $1/(1 - \beta)$, and $\mathcal{L}^{-1}(\pm \xi e) = \pm \xi/(1 \pm \beta)^2$, which makes the asymmetry of the inverse Legendre transform explicit.

Obstacles enter as a conformal factor $s(q)F$ with $s \geq 1$; the scaled metric is again Randers with $A \mapsto s^2 A$, $b \mapsto sb$, its dual norm is F^*/s , and its fundamental tensor is $s^2 g_v$.

B. Sum and p -sum families

For $F(q, v) = \sum_{k=1}^K \|L_k(q)v\| + b(q)^\top v$ with $b = \rho L_1^\top u$, $\rho \in [0, 1)$, $\|u\| \leq 1$, positivity follows from $b^\top v \geq -\rho\|u\| \|L_1 v\| > -\|L_1 v\|$ and strong convexity from $\partial_{vv}^2 F \geq \partial_{vv}^2 \|L_1 v\| > 0$ on hyperplanes transverse to v . The fundamental tensor is closed form,

$$g_v = dF dF^\top + F \sum_k \left(\frac{A_k}{\alpha_k} - \frac{(A_k v)(A_k v)^\top}{\alpha_k^3} \right), \quad (3)$$

$$A_k = L_k^\top L_k, \quad \alpha_k = \|L_k v\|,$$

the inverse Legendre transform is computed by a damped Newton iteration with Jacobian g_v , and geodesics are integrated in Euler–Lagrange form $g_v \dot{v} = \partial_q E - (\partial_q p)v$ with $E = F^2/2$, $p = \partial_v E$, where $\partial_q E$ is obtained by reverse-mode and $(\partial_q p)v$

by forward-mode automatic differentiation. For $K = 1$ the family is Randers and the Euler–Lagrange integrator agrees with the Hamiltonian integrator to 10^{-10} . The p -sum family $(\sum_k \|L_k v\|^p)^{1/p} + b^\top v$ adds a learned exponent $p \geq 2$ and a small regularizing ellipsoid so that strong convexity is retained near the flat faces; p is fitted jointly with L_k and b .

C. Left-invariant metrics on SE(2)

Writing $q = (x, y, \theta)$, $\Lambda(\theta) = \text{diag}(R(\theta), 1)$ and the body velocity $\xi = \Lambda(\theta)^{-1} \dot{q} = (v_x, v_y, \omega)$, a body-frame envelope F_{body} defines the left-invariant metric $F(q, \dot{q}) = F_{\text{body}}(\xi)$. Its geodesics satisfy the Euler–Poincaré equations in the body momentum $p = \mathcal{L}_{\text{body}}^{-1}(\xi)$,

$$\begin{aligned} \dot{p} &= \text{ad}_\xi^* p = (\omega p_y, -\omega p_x, v_y p_x - v_x p_y), \\ \dot{q} &= \Lambda(\theta) \mathcal{L}_{\text{body}}^{-1}(p), \end{aligned} \quad (4)$$

which require no spatial derivative of F . For a Randers body metric $\mathcal{L}_{\text{body}}^{-1}$ is closed form and the integrator runs in NumPy; for the p -sum family the Newton inverse is warm-started along the trajectory. The quadruped-type envelope of the paper (forward 1.2, backward 0.5, lateral 0.4 m/s, yaw 1.2 rad/s) corresponds to $A = \text{diag}(a_x, a_y, a_\theta)$ and $b = (b_x, 0, 0)$ with $a_x^{1/2} + b_x = 1/1.2$ and $a_x^{1/2} - b_x = 1/0.5$; its symmetric part has fore–aft speed $1/a_x^{1/2} = 0.71$ m/s. A physical check: for a goal 1.5 m directly behind the robot with the same heading, the minimizing geodesic backs up in 3.0 s, and with reversed heading it turns and walks in 3.35 s.

II. PROOFS

A. Forward geodesic velocity (Proposition 1)

Let $\rho(x) = d_F(x, x_1)$. Since $\rho(x) = d_{\bar{F}}(x_1, x)$, ρ is the distance from x_1 in the reverse metric $\bar{F}$. By the standard result for distance functions on Finsler manifolds [6], wherever ρ is differentiable its $\bar{F}$ -gradient $\bar{\mathcal{L}}^{-1}(d\rho)$ is the arrival velocity u of the unit-speed $\bar{F}$ -geodesic from x_1 to x . Reversing a $\bar{F}$ -geodesic yields an F -geodesic, so the minimizing F -geodesic from x to x_1 starts with $v = -u$. Because $\bar{g}_u = g_{-u}$, we have $\bar{\mathcal{L}}(u) = g_{-u}(u, \cdot) = -\mathcal{L}(-u)$, hence $\bar{\mathcal{L}}(u) = d\rho$ is equivalent to $\mathcal{L}(-u) = -d\rho$, i.e. $v = \mathcal{L}^{-1}(-d\rho)$. The eikonal equation $F^*(-d\rho) = 1$ gives $F(v) = F^*(-d\rho) = 1$ and $d\rho(v) = -g_v(v, v) = -1$. $\square$

The negative gradient $-\nabla^F \rho = -\mathcal{L}^{-1}(d\rho)$ differs from v in direction and magnitude; for the constant Randers metric $F = \|v\| + \beta v_1$ shooting along it from a generic point misses x_1 by a distance that grows with β .

B. Conditional target field and schedule

Let x_t move along the minimizing geodesic from x_0 to x_1 with $\rho(x_t) = \kappa(t)\rho(x_0)$, κ decreasing, $\kappa(0) = 1$, $\kappa(1) = 0$. By Proposition 1 the velocity is $u_t = c(t, x)v(x)$ with $F(v) = 1$ and $d\rho(v) = -1$, so $\frac{d}{dt}\rho(x_t) = d\rho(u_t) = -c$. Differentiating the schedule gives $-c = \dot{\kappa}\rho(x_0) = (\dot{\kappa}/\kappa)\rho(x_t)$, hence

$$u_t(x | x_1) = -\frac{\dot{\kappa}(t)}{\kappa(t)}\rho(x)\mathcal{L}_x^{-1}(-d\rho_x), \quad (5)$$

which for $\kappa = 1 - t$ is the expression in the paper. For a constant Minkowski metric $\rho(x) = F(x_1 - x)$ and $\mathcal{L}^{-1}(-d\rho) = (x_1 - x)/F(x_1 - x)$, so $u_t = (x_1 - x)/(1 - t)$. $\square$

C. Admissible loss weights (Remark 1)

Write the loss as $\mathbb{E}_{t, x_t}[\mathbb{E}_z[(v_\theta - u_t)^\top M(v_\theta - u_t) | t, x_t]]$. For fixed (t, x_t) the inner expectation is a quadratic in the vector $v_\theta(t, x_t)$ with Hessian $2\mathbb{E}[M | t, x_t]$ and stationary point $(\mathbb{E}[M | t, x_t])^{-1}\mathbb{E}[Mu_t | t, x_t]$. If M is $\sigma(t, x_t)$ -measurable it factors out of both expectations and the minimizer is $\mathbb{E}[u_t | t, x_t]$, independent of M ; this is the marginal vector field, which satisfies the continuity equation of the mixture of conditional paths [5], so its flow transports p_0 to p_1 . If M depends on z the minimizer is the M -weighted conditional mean, which equals the marginal field only when $\text{Cov}(M, u_t | t, x_t) = 0$. $\square$

In the Riemannian case $g_v \equiv g(x)$ is $\sigma(t, x_t)$ -measurable, so the metric-weighted loss of Riemannian flow matching is admissible; the distortion is specific to velocity-dependent fundamental tensors.

D. Closed drifts do not change paths

If $F = \alpha + \beta$ with a Riemannian part α and a closed one-form $\beta = d\varphi$ (for instance a slope model $b \propto \nabla h$), then for every curve $L_F(\gamma) = L_\alpha(\gamma) + \varphi(x_1) - \varphi(x_0)$, so the minimizers coincide with those of α and only travel times change. Direction-dependent paths require a drift with non-vanishing curl (shear, vortex, channel) or a pose-dependent body envelope, which is why the experiments use those.

III. LOSS-WEIGHT EXPERIMENT

The constant Randers metric $F = \|v\| + \beta v_1$ (slow direction $+x$), a three-mode Gaussian-mixture target with equal weights, and a standard normal source define a conditional flow-matching problem whose conditional paths are straight lines for every loss weight. We compute the exact population minimizers for $M = I$ and for $M = g_{u_t}(x_t | x_1)$, integrate both flows, and compare the generated samples with fresh target samples by the energy distance (Table I, Fig. 1). The (t, x) -dependent weight recovers the target for every β ; the endpoint-dependent weight favors targets in the slow direction by $(1 + \beta)^2/(1 - \beta)^2$ and collapses the mass onto that mode.

TABLE I
ENERGY DISTANCE TO FRESH TARGET SAMPLES (TWO FRESH SAMPLES DIFFER BY 0.001) AND MASS OF THE THREE MODES (LEFT / RIGHT / TOP).

β	$M = I$ (any $M(t, x)$)	$M = g_{u_t}$ (x_1 -dependent)
0.0	0.002, mass 0.31 / 0.34 / 0.35	identical
0.3	0.002, mass 0.31 / 0.34 / 0.35	0.348, mass 0.14 / 0.57 / 0.29
0.6	0.002, mass 0.31 / 0.34 / 0.35	0.925, mass 0.05 / 0.74 / 0.21
0.9	0.002, mass 0.31 / 0.34 / 0.35	1.283, mass 0.02 / 0.83 / 0.15

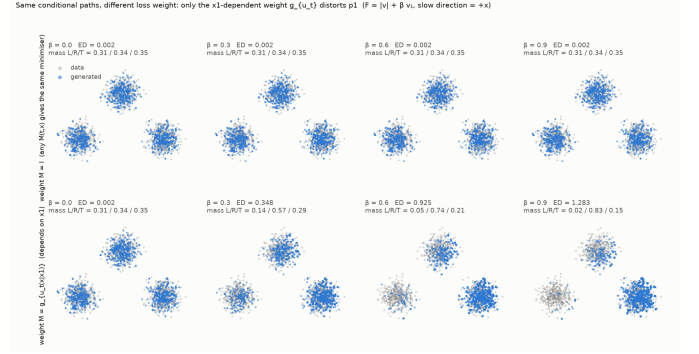


Fig. 1. Exact population minimizers of the flow-matching loss under the two weights, integrated from the source distribution.

IV. BOUNDARY-VALUE-FREE CONSTRUCTION

For a fixed endpoint x_1 and $w \sim \mathcal{N}(0, \sigma^2 I)$ in $T_{x_1}\mathcal{M}$, set $x_0 = \text{Exp}_{x_1}^{\bar{F}}(w)$. By homogeneity of the geodesic spray, $x_t = \text{Exp}_{x_1}^{\bar{F}}((1 - t)w)$ and the conditional target is $u_t = -\frac{d}{ds}\text{Exp}_{x_1}^{\bar{F}}(sw)|_{s=1-t}$, the reversed $\bar{F}$ -velocity, which equals the F -geodesic velocity at the same point. This avoids ρ , $d\rho$ and the Legendre pitfall altogether; the price is that the source distribution is the target diffused along reverse geodesics rather than a fixed Gaussian, and at inference x_0 is drawn by the same procedure from a discarded endpoint. We use this construction for the demonstration-tracking experiment of Sec. XI, where only the demonstration distribution is prescribed.

V. TRAINING DATA AND HYPERPARAMETERS

TABLE II
DATA GENERATION AND TRAINING SETTINGS.

Setting	Plane	SE(2)	Obstacles	β -cond.
Sampled pairs	2000	3000	1600	2400
Retained (Finsler)	1324	1581	1600	2107
Branches	3	3 wraps	grid + polish	3
Minimality tolerance	5%	10%	—	5%
Samples per geodesic	48	48	48	48
Inputs	t, x, g	t, q (θ as $\cos, \sin$)	t, x, g	t, x, g, β
MLP	4×256	4×256	4×256	4×256
Steps / batch / lr	6k / 2048 / 10^{-3}	8k / 2048 / 10^{-3}	6k / 2048 / 10^{-3}	6k / 2048 / 10^{-3}
Evaluation queries	98	192	98	3×98

The shooting solver is a Gauss–Newton iteration on the initial momentum with per-sample backtracking, continuation

in the asymmetry strength (from $b = 0$ to the target b) and, in the plane, continuation in the target point; the three initial branches are the straight connection and detours above and below the channel (on SE(2), three heading wraps). The obstacle conditional paths are obtained by grid search selecting the homotopy class, followed by 400 Adam steps on the interior nodes of the polyline minimizing its discrete Finsler length with a small smoothness term, and resampled at constant Finsler speed; polished paths are 0.3–1.6% shorter than the grid paths and collision free.

VI. GRID REFERENCE CONVERGENCE

TABLE III
RELATIVE ERROR OF THE GRID SEARCH TIME AGAINST
BOUNDARY-VALUE-VERIFIED MINIMIZING GEODESICS (59 PLANAR PAIRS,
60 SE(2) PAIRS).

Space	h (angle)	Nodes	Edges	Mean error	Max
Plane	0.08	6.7k	0.2M	+0.37%	2.5%
Plane	0.04	27k	0.8M	+0.38%	1.2%
Plane	0.02	106k	3.4M	+0.40%	0.9%
SE(2)	0.14	51k	4.7M	+2.9%	20%
	(15°)				
SE(2)	0.10	152k	14.4M	+3.1%	11%
	(10°)				
SE(2)	0.07	406k	38.8M	+2.1%	12%
	(7.5°)				

The grid overestimates the true time by the angular quantization of its 32 (plane) or 98 (SE(2)) edge directions plus midpoint quadrature; the SE(2) maxima are pairs on which the shooting solver sat on a non-minimal branch (grid shorter than the solver), which the minimality filter excludes from training.

VII. SCORING SE(2) PLANS AGAINST EXACT GEODESICS

The grid reference of the main paper is an upper bound on the true time, so a ratio below one cannot by itself establish optimality. Table IV re-scores the stored SE(2) plans two further ways: against a finer grid ($h = 0.07$ m, 7.5° , whose overestimate against verified geodesics is +2.1% instead of +3.1%), and against exact shooting geodesics on the 98 of 192 poses where the solver converges and the grid certifies the solution minimal. The three rankings agree, and against the exact optimum the Finsler planner is within 0.3% of optimal; the remaining gap below one is the 0.2 m position tolerance, which lets a plan stop marginally short of the goal.

TABLE IV
TRAVEL RATIO OF THE SAME STORED PLANS UNDER THREE REFERENCES.
THE EXACT COLUMN IS RESTRICTED TO THE 98 POSES WITH A CERTIFIED
MINIMAL GEODESIC, WHICH IS WHY EVERY METHOD IMPROVES
SLIGHTLY.

Conditional path	Grid $h=0.1$	Grid $h=0.07$	Exact geodesic
Euclidean	1.254	1.254	1.212
Riemannian	1.073	1.073	1.064
Finsler	0.979	0.979	0.998

VIII. DOES THE GENERATIVE MECHANISM MATTER?

The main paper compares conditional-path geometries inside one flow-matching framework. The complementary question is whether the mechanism that turns those paths into a planner matters, so we train two further learners on exactly the same 1581 Finsler geodesics and score them on exactly the same 192 held-out poses (Table V). The trajectory regressor maps the start pose to all 48 waypoints in one forward pass. The trajectory diffusion model is a denoising diffusion model over the whole trajectory in the style of Janner et al. [3] and Carvalho et al. [2]: a temporal convolutional backbone along the waypoint axis, the start pose and the diffusion step injected per channel, 100 denoising steps, and the first and last waypoints pinned by inpainting. Both baselines are trained for 30k steps against the 8k steps of our flow.

TABLE V
LEARNERS TRAINED ON THE SAME FINSLER GEODESICS, SAME
EVALUATION POSES. ALL THREE INHERIT THE GEOMETRY AND LAND
WITHIN TWO PER CENT OF THE REFERENCE; THE FLOW IS THE MOST
ACCURATE AND, UNLIKE THE DIFFUSION SAMPLER, NEEDS ONE ODE
SOLVE RATHER THAN 100 DENOISING STEPS.

Learner	Ratio	Median	Worst	Latency
Trajectory regression	1.010	0.986	1.60	0.06 ms
Trajectory diffusion	0.996	0.986	1.35	4.58 ms
Finsler flow matching	0.979	0.972	1.26	0.64 ms

The margins are small but consistent rather than noise: paired over the 192 poses, the flow beats the regressor by 0.031 ± 0.005 (6.1 standard errors, better on 99% of poses) and the diffusion planner by 0.017 ± 0.003 (5.2 standard errors, better on 95%).

Read together with the path-family floors of the main paper (1.205 Euclidean, 1.058 symmetric Riemannian, 1.000 Finsler, before any learning), the picture is that the geometry sets the achievable time and the learner recovers it to within a few per cent. This is the intended reading of the paper: the contribution is the geometry, and the flow is the instantiation that is both the most accurate of the three and the one whose output is a field rather than a trajectory, which is what makes the closed-loop use of Sec. IX possible from the same model.

IX. FULL RESULT TABLES

TABLE VI
PLANAR DRIFT FIELD, 98 HELD-OUT QUERIES. TRAVEL TIME OVER
OPTIMAL (MEAN $\pm$ S.E.), MEDIAN, WORST, REACH RATE WITHIN 0.35,
AND TIME-TO-GOAL RATIO INCLUDING THE OPTIMAL REMAINING LEG.

Path	β	Mean $\pm$ s.e. (median)	Worst	Reach	Total
Euclidean	0.9	2.551 ± 0.144 (2.10)	6.32	0.99	2.58
Riemannian	0.9	1.082 ± 0.005 (1.08)	1.21	0.89	1.15
Finsler	0.9	1.000 ± 0.001 (0.998)	1.02	0.89	1.06
Euclidean	0.7	1.465 ± 0.036	2.29	0.98	1.48
Riemannian	0.7	1.055 ± 0.003	1.11	0.95	1.09
Finsler	0.7	0.996 ± 0.000	1.01	0.90	1.03

In the plane the Euclidean $t = 0$ field succeeds on 37–100% of runs at 1.4–2.6 times the optimum, and the Finsler

TABLE VII
PLANNER WITH THREE OBSTACLES, $\beta = 0.9$, 98 QUERIES; ALL
CONDITIONAL PATHS ARE OBSTACLE-AWARE.

Path	Reach	Collision	Median (mean)	Worst	Total median
Euclidean	0.84	0.00	2.97 (2.62)	4.99	2.99
Riemannian	0.71	0.05	1.064 (1.14)	3.29	1.10
Finsler	0.91	0.00	0.999 (1.001)	1.05	1.018

TABLE VIII
ONE β -CONDITIONED PLANNER TRAINED ON $\beta \in [0.30, 0.85]$; MEAN
RATIO (WORST) OVER 98 QUERIES.

Test β	Euclidean	Riemannian	Finsler	Finsler total
0.50	1.187 (1.47)	1.046 (1.14)	0.997 (1.01)	1.025
0.90	2.513 (6.56)	1.073 (1.16)	0.999 (1.05)	1.052
0.95	3.684 (12.95)	1.079 (1.21)	1.000 (1.10)	1.065

$t = 0$ field, trained from start samples only, on 67–100%; the streaming clock uses the $t > 0$ field, which is trained on every point of every path, and restores success on every run. As a feedback law the streaming Finsler flow, direct regression and the reference are statistically tied. The flow’s role is that the single model covers both regimes: the field that generates the open-loop plan, with its travel-time estimate, is the same field that is re-evaluated as the feedback law, so no second model is trained and replanning costs one network evaluation.

X. H1-2 SIMULATION DETAILS

The environment is MuJoCo 3.13 with the 12-degree-of-freedom H1-2 model and the pretrained velocity-command policy of `unitree_rl_gym` [7] (47-dimensional observation: scaled angular velocity, projected gravity, scaled command, joint positions relative to default, scaled joint velocities, last action, gait phase; 50 Hz policy, PD joint control at 500 Hz). Commands are clipped to the asymmetric interface limits before entering the policy. Envelope identification sweeps 80 directions of the scaled command sphere (scales 1.6, 1.0, 2.0 for v_x , v_y , ω) at 5 magnitudes: each rollout resets the robot standing, holds a zero command for 1 s, then the test command for 5 s, and records the mean body-frame velocity over the last 3 s; a rollout counts as a failure if the base height drops below 0.55 m or the projected gravity indicates a fall. The directional maxima of the achieved velocities are fitted by a Randers metric in the body frame; the symmetric baseline is its Riemannian part. For execution, an SE(2) value function is computed for each metric, and every 50 ms the descent direction is converted into a body-velocity command at 90% of the envelope speed in that direction; a task succeeds when the pose is within 0.25 m and 15° of the goal. The planner’s own travel-time estimates are 3.6, 2.9 and 2.1 s for the three asymmetric tasks against executed 4.8, 4.0 and 3.0 s, a uniform execution lag of the locomotion policy that does not affect the

TABLE IX
ENVELOPE IDENTIFICATION ON SE(2): FITTED AXIS SPEEDS (FORWARD /
BACKWARD / LATERAL [M/S], YAW [RAD/S]), RADIAL ERROR OF THE
INDICATRIX, AND EXECUTED TIME OVER THE TRUE OPTIMUM (96 POSES).

Family	Speeds	Err. mean / max	Ratio mean / median / worst
True	1.20 / 0.50 / 0.40 / 1.20	0	1.000
Randers ($K = 1$)	1.22 / 0.61 / 0.45 / 1.45	9.0% / 28%	1.036 / 1.031 / 1.136
Sum ($K = 2$)	1.27 / 0.63 / 0.43 / 1.31	7.8% / 25%	1.027 / 1.018 / 1.135
p -sum ($K = 3$, $p = 5.3$)	1.16 / 0.60 / 0.41 / 1.37	7.2% / 19%	1.023 / 1.016 / 1.116
Symmetric part	0.82 / 0.82 / 0.45 / 1.44	15.1% / 63%	1.126 / 1.059 / 1.744

TABLE X
PER-QUERY LATENCY ON ONE CPU (WALL-CLOCK MEDIANS).

Space	Method	Single query	Batch of 100
Plane	FM plan, 50 RK4 steps	13.2 ms	0.69 ms
Plane	FM plan, 200 RK4 steps	50.6 ms	3.40 ms
Plane	FM feedback field	3 μ s	—
Plane	Grid search ($h = 0.04$, 32 dirs)	11 ms per goal	+0.14 s build, 0.8M edges
Plane	Shooting BVP	22.1 s	2.24 s
SE(2)	FM plan, 50 RK4 steps	28.8 ms	1.54 ms
SE(2)	FM plan, 200 RK4 steps	130 ms	6.16 ms
SE(2)	Grid search ($h = 0.1$, 10° , 98 dirs)	330 ms per goal	+9.9 s build, 14.4M edges
SE(2)	Shooting BVP	74.5 s	—

comparison. Fig. 2 gives the executed paths of all four tasks seen from above.

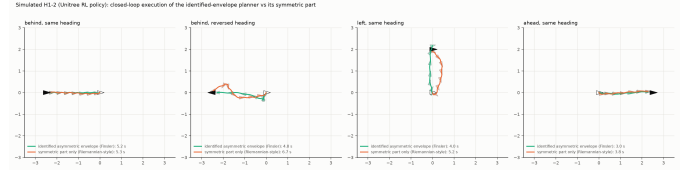


Fig. 2. Executed paths on the simulated H1-2 for the four tasks, planner using the identified asymmetric envelope against its symmetric part.

One caution for reproduction: the pretrained locomotion policy carries state between calls, so arrival times depend on the order in which the tasks are run. All numbers reported here come from the single task order of the demo script, and the snapshots of the main paper replay that same order.

XI. STABILIZING TARGETS FOR DEMONSTRATION TRACKING

In the streaming-policy setting [4] the target field for tracking a demonstration $\xi(t)$ is $\hat{\xi}(t) + k \text{Log}_a(\xi(t))$, the demonstration velocity plus a stabilizing pull towards it. Replacing the Euclidean log by the Finsler log of Proposition

TABLE XI

CLOSED-LOOP EXECUTION, TIME-TO-GOAL OVER THE UNDISTURBED OPTIMAL TIME (SUCCESS RATE IN PARENTHESES WHERE BELOW ONE). SE(2): 192 POSES, TOLERANCE $0.15 \text{ m} / 10^\circ$, $t = 0$ FEEDBACK FIELDS. PLANE: 49 STARTS TO THE CORE GOAL, TOLERANCE 0.3; THE MODEL IS TRAINED AT $\beta = 0.9$ AND GUSTS AND LULLS CHANGE THE TRUE WIND; BC IS DIRECT REGRESSION OF THE OPTIMAL VELOCITY DIRECTION ON THE SAME TRAINING PATHS.

Space	Disturbance	Euclidean	Symmetric	Finsler	Reference
SE(2)	none	1.175	1.017	0.931	0.953
SE(2)	slip	1.198	1.042	0.951	0.977
SE(2)	slip + drift	1.339	1.114	0.989	1.028
SE(2)	slip + pushes	1.256	1.097	0.998	1.032
		Riem. $t=0$	Finsler stream	BC	Reference
Plane	none	1.28	0.97	0.96	0.95
Plane	slip	1.03	0.99	0.97	0.96
Plane	gust +30%	1.18	1.00	0.98	0.96
Plane	lull -30%	0.98	0.99	1.00	0.98
Plane	slip + pushes	1.09	1.01	1.00	0.98

1, $\text{Log}_a^F(y) = \rho_y(a) \mathcal{L}_a^{-1}(-d\rho_y)$, makes the stabilizing term the forward geodesic direction; by subadditivity of F the commanded speed satisfies $F(a, v) \leq F(a, \dot{\xi}) + k\rho_{\xi(t)}(a)$, so envelope violations near an in-envelope demonstration are bounded by $k\rho$. Table XII reports 48 time-optimal demonstrations in the $\beta = 0.9$ channel, tracked from starts perturbed by $|\delta|$ in eight directions (384 rollouts per condition). Near the demonstrations the three log maps coincide; at $|\delta| = 0.8$ the Finsler log reduces both the envelope violation and the projected tracking error by 13%, the Riemannian log gives no gain over the Euclidean one, and a network-free oracle reproduces the learned numbers, so the gain is geometric.

TABLE XII

DEMONSTRATION TRACKING: MEAN ENVELOPE VIOLATION / PROJECTED TRACKING ERROR AT $t = 1$ FOR PERTURBED STARTS.

$ \delta $	Euclidean log	Riemannian log	Finsler log
0.1	0.021 / 0.043	0.022 / 0.043	0.022 / 0.042
0.25	0.057 / 0.114	0.057 / 0.113	0.056 / 0.110
0.5	0.132 / 0.264	0.132 / 0.259	0.124 / 0.241
0.8	0.258 / 0.490	0.258 / 0.483	0.225 / 0.425

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